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Last login: Sun Jun 28 06:53:48 on ttys000
stephenrajva@Stephens-MacBook-Air-4 ~ % cd "/Users/stephenrajva/Downloads/HPC-Up
load 2/SST_Stimulator/MiniMD/Torus/PPN_2"
chmod +x run_minimd_torus_sweep_ppn2.sh
export MESH_SHAPE="2x2x2"; export STATS_DIR="./results"
sst minimd_torus3d_ppn2.py
```

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=====
Topology          : 3D Torus 2x2x2 - MiniMD
Routers           : 8
Nodes per router  : 2 (local_ports)
Total nodes       : 16
Total MPI ranks   : 16
PPN (ranks/node)  : 2 <-- should be 2
Stats CSV         : ./results/minimd_torus_2x2x2_ppn2_nodes16_stats.csv
=====
```

```
180000:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
180000:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
180000:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
180000:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
180000:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
180000:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
180000:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
180000:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
162250536921:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
162250536921:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
162250536921:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
162250536921:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
162251053179:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
162251053179:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
162251053179:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
162251053179:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
AllPingPong: total time 162253.347 us, loop 325, bufLen 20510, latency 249.621 u
s. bandwidth 0.082165 GB/s
162253527383:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
162253527383:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
162253527383:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
162253527383:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
162254043257:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
162254043257:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
162254043257:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
162254043257:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
318058829298:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
318058829298:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
318058829298:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
318058829298:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
318059413206:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
318059413206:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
318059413206:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
318059413206:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
AllPingPong: total time 155806.815 us, loop 325, bufLen 9000, latency 239.703 us
```

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. bandwidth 0.037546 GB/s
318060342149:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
318060342149:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
318060342149:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
318060342149:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
318060926057:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
318060926057:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
318060926057:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
318060926057:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
480310187271:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
480310187271:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
480310187271:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
480310187271:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
480310703529:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
480310703529:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
480310703529:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
480310703529:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
AllPingPong: total time 162252.835 us, loop 325, bufLen 20510, latency 249.620 u
s. bandwidth 0.082165 GB/s
480313177733:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
480313177733:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
480313177733:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
480313177733:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
480313693607:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
480313693607:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
480313693607:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
480313693607:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
AllPingPong: total time 155806.815 us, loop 325, bufLen 9000, latency 239.703 us
. bandwidth 0.037546 GB/s
Allreduce: ranks 16, loop 40, 1 double(s), latency 2457.964 us
Barrier: ranks 16, loop 15, latency 1399.044 us
Simulation is complete, simulated time: 755.434 ms
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HP
C-Upload 2/SST_Stimulator/MiniMD/Torus/PPN_2"
./run_minimd_torus_sweep_ppn2.sh
ls -l results/*.csv
=== MiniMD 3D torus 2x2x2 (PPN=2) ===
=====
Topology          : 3D Torus 2x2x2 - MiniMD
Routers           : 8
Nodes per router  : 2 (local_ports)
Total nodes       : 16
Total MPI ranks   : 16
PPN (ranks/node)  : 2 <-- should be 2
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/M
iniMD/Torus/PPN_2/results/minimd_torus_2x2x2_ppn2_nodes16_stats.csv
=====
180000:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
180000:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
180000:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
180000:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
180000:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
180000:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
180000:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
180000:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16

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180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
162250536921:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
162250536921:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
162250536921:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
162250536921:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
162251053179:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
162251053179:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
162251053179:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
162251053179:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
AllPingPong: total time 162253.347 us, loop 325, bufLen 20510, latency 249.621 u
s. bandwidth 0.082165 GB/s
162253527383:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
162253527383:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
162253527383:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
162253527383:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
162254043257:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
162254043257:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
162254043257:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
162254043257:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
318058829298:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
318058829298:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
318058829298:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
318058829298:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
318059413206:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
318059413206:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
318059413206:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
318059413206:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
AllPingPong: total time 155806.815 us, loop 325, bufLen 9000, latency 239.703 us
. bandwidth 0.037546 GB/s
318060342149:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
318060342149:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
318060342149:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
318060342149:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
318060926057:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
318060926057:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
318060926057:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
318060926057:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
480310187271:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
480310187271:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
480310187271:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
480310187271:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
480310703529:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
480310703529:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
480310703529:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
480310703529:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
AllPingPong: total time 162252.835 us, loop 325, bufLen 20510, latency 249.620 u
s. bandwidth 0.082165 GB/s
480313177733:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
480313177733:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
480313177733:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
480313177733:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
480313693607:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
480313693607:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
480313693607:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16

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480313693607:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
AllPingPong: total time 155806.815 us, loop 325, bufLen 9000, latency 239.703 us
. bandwidth 0.037546 GB/s
Allreduce: ranks 16, loop 40, 1 double(s), latency 2457.964 us
Barrier: ranks 16, loop 15, latency 1399.044 us
Simulation is complete, simulated time: 755.434 ms
=== MiniMD 3D torus 2x2x4 (PPN=2) ===
=====
Topology          : 3D Torus 2x2x4 - MiniMD
Routers           : 16
Nodes per router  : 2 (local_ports)
Total nodes       : 32
Total MPI ranks   : 32
PPN (ranks/node)  : 2  <-- should be 2
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/M
iniMD/Torus/PPN_2/results/minimd_torus_2x2x4_ppn2_nodes32_stats.csv
=====
180000:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=32
180000:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=32
180000:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=32
180000:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=32
180000:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=32
180000:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=32
180000:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=32
180000:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=32
180000:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=32
180000:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=32
180000:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=32
180000:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=32
180000:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=32
180000:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=32
180000:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=32
180000:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=32
180000:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=32
180000:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=32
180000:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=32
180000:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=32
180000:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=32
180000:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=32
180000:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=32
180000:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=32
180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=32
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=32
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=32
180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=32
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=32
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=32
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=32
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=32
162382214727:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=32
162382214727:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=32
162382214727:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=32
162382214727:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=32
162382730985:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=32
162382730985:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=32
162382730985:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=32
162382730985:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=32
162385138489:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=32

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162385138489:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=32
162385138489:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=32
162385138489:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=32
162385654363:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=32
162385654363:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=32
162385654363:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=32
162385654363:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=32
162386991781:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=32
162386991781:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=32
162386991781:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=32
162386991781:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=32
162387508039:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=32
162387508039:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=32
162387508039:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=32
162387508039:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=32
AllPingPong: total time 162389.735 us, loop 325, bufLen 20510, latency 249.830 u
s. bandwidth 0.082096 GB/s
162389915543:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=32
162389915543:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=32
162389915543:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=32
162389915543:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=32
162390431417:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=32
162390431417:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=32
162390431417:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=32
162390431417:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=32
318233728704:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=32
318233728704:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=32
318233728704:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=32
318233728704:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=32
318234312612:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=32
318234312612:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=32
318234312612:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=32
318234312612:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=32
318235308255:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=32
318235308255:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=32
318235308255:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=32
318235308255:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=32
318235892163:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=32
318235892163:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=32
318235892163:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=32
318235892163:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=32
318238505758:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=32
318238505758:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=32
318238505758:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=32
318238505758:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=32
318239089666:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=32
318239089666:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=32
318239089666:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=32
318239089666:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=32
AllPingPong: total time 155850.170 us, loop 325, bufLen 9000, latency 239.769 us
. bandwidth 0.037536 GB/s
318240085309:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=32
318240085309:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=32
318240085309:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=32
318240085309:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=32
318240669217:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=32
318240669217:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=32
318240669217:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=32

```

318240669217:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=32
480615285077:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=32
480615285077:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=32
480615285077:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=32
480615285077:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=32
480615801335:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=32
480615801335:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=32
480615801335:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=32
480615801335:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=32
480618208839:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=32
480618208839:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=32
480618208839:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=32
480618208839:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=32
480618724713:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=32
480618724713:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=32
480618724713:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=32
480618724713:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=32
480620062131:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=32
480620062131:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=32
480620062131:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=32
480620062131:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=32
480620578389:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=32
480620578389:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=32
480620578389:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=32
480620578389:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=32
AllPingPong: total time 162382.900 us, loop 325, bufLen 20510, latency 249.820 u
s. bandwidth 0.082099 GB/s
480622985893:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=32
480622985893:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=32
480622985893:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=32
480622985893:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=32
480623501767:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=32
480623501767:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=32
480623501767:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=32
480623501767:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=32
AllPingPong: total time 155850.170 us, loop 325, bufLen 9000, latency 239.769 us
. bandwidth 0.037536 GB/s
Allreduce: ranks 32, loop 40, 1 double(s), latency 2460.022 us
Barrier: ranks 32, loop 15, latency 1401.150 us
Simulation is complete, simulated time: 755.905 ms
=== MiniMD 3D torus 4x4x4 (PPN=2) ===
=====
Topology          : 3D Torus 4x4x4 - MiniMD
Routers           : 64
Nodes per router  : 2 (local_ports)
Total nodes       : 128
Total MPI ranks   : 128
PPN (ranks/node)  : 2  <-- should be 2
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/M
iniMD/Torus/PPN_2/results/minimd_torus_4x4x4_ppn2_nodes128_stats.csv
=====
180000:0:127:EmberEngine:AllPingPong:generate:59: rank=127 size=128
180000:0:126:EmberEngine:AllPingPong:generate:59: rank=126 size=128
180000:0:125:EmberEngine:AllPingPong:generate:59: rank=125 size=128
180000:0:124:EmberEngine:AllPingPong:generate:59: rank=124 size=128
180000:0:123:EmberEngine:AllPingPong:generate:59: rank=123 size=128
180000:0:122:EmberEngine:AllPingPong:generate:59: rank=122 size=128
180000:0:121:EmberEngine:AllPingPong:generate:59: rank=121 size=128

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162385654363:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=128
162385654363:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=128
162385654363:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=128
162385654363:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=128
162385654363:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=128
162385654363:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=128
162385654363:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=128
162385654363:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=128
162385654363:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=128
162385654363:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=128
162385654363:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=128
162386991781:0:64:EmberEngine:AllPingPong:generate:59: rank=64 size=128
162386991781:0:66:EmberEngine:AllPingPong:generate:59: rank=66 size=128
162386991781:0:68:EmberEngine:AllPingPong:generate:59: rank=68 size=128
162386991781:0:70:EmberEngine:AllPingPong:generate:59: rank=70 size=128
162386991781:0:72:EmberEngine:AllPingPong:generate:59: rank=72 size=128
162386991781:0:74:EmberEngine:AllPingPong:generate:59: rank=74 size=128
162386991781:0:76:EmberEngine:AllPingPong:generate:59: rank=76 size=128
162386991781:0:78:EmberEngine:AllPingPong:generate:59: rank=78 size=128
162386991781:0:80:EmberEngine:AllPingPong:generate:59: rank=80 size=128
162386991781:0:82:EmberEngine:AllPingPong:generate:59: rank=82 size=128
162386991781:0:84:EmberEngine:AllPingPong:generate:59: rank=84 size=128
162386991781:0:86:EmberEngine:AllPingPong:generate:59: rank=86 size=128
162386991781:0:88:EmberEngine:AllPingPong:generate:59: rank=88 size=128
162386991781:0:90:EmberEngine:AllPingPong:generate:59: rank=90 size=128
162386991781:0:92:EmberEngine:AllPingPong:generate:59: rank=92 size=128
162386991781:0:94:EmberEngine:AllPingPong:generate:59: rank=94 size=128
162387508039:0:65:EmberEngine:AllPingPong:generate:59: rank=65 size=128
162387508039:0:67:EmberEngine:AllPingPong:generate:59: rank=67 size=128
162387508039:0:69:EmberEngine:AllPingPong:generate:59: rank=69 size=128
162387508039:0:71:EmberEngine:AllPingPong:generate:59: rank=71 size=128
162387508039:0:73:EmberEngine:AllPingPong:generate:59: rank=73 size=128
162387508039:0:75:EmberEngine:AllPingPong:generate:59: rank=75 size=128
162387508039:0:77:EmberEngine:AllPingPong:generate:59: rank=77 size=128
162387508039:0:79:EmberEngine:AllPingPong:generate:59: rank=79 size=128
162387508039:0:81:EmberEngine:AllPingPong:generate:59: rank=81 size=128
162387508039:0:83:EmberEngine:AllPingPong:generate:59: rank=83 size=128
162387508039:0:85:EmberEngine:AllPingPong:generate:59: rank=85 size=128
162387508039:0:87:EmberEngine:AllPingPong:generate:59: rank=87 size=128
162387508039:0:89:EmberEngine:AllPingPong:generate:59: rank=89 size=128
162387508039:0:91:EmberEngine:AllPingPong:generate:59: rank=91 size=128
162387508039:0:93:EmberEngine:AllPingPong:generate:59: rank=93 size=128
162387508039:0:95:EmberEngine:AllPingPong:generate:59: rank=95 size=128
AllPingPong: total time 162389.735 us, loop 325, bufLen 20510, latency 249.830 u
s. bandwidth 0.082096 GB/s
162389915543:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=128
162389915543:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=128
162389915543:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=128
162389915543:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=128
162389915543:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=128
162389915543:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=128
162389915543:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=128
162389915543:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=128
162389915543:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=128
162389915543:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=128
162389915543:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=128
162389915543:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=128
162389915543:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=128

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480620578389:0:71:EmberEngine:AllPingPong:generate:59: rank=71 size=128
480620578389:0:73:EmberEngine:AllPingPong:generate:59: rank=73 size=128
480620578389:0:75:EmberEngine:AllPingPong:generate:59: rank=75 size=128
480620578389:0:77:EmberEngine:AllPingPong:generate:59: rank=77 size=128
480620578389:0:79:EmberEngine:AllPingPong:generate:59: rank=79 size=128
480620578389:0:81:EmberEngine:AllPingPong:generate:59: rank=81 size=128
480620578389:0:83:EmberEngine:AllPingPong:generate:59: rank=83 size=128
480620578389:0:85:EmberEngine:AllPingPong:generate:59: rank=85 size=128
480620578389:0:87:EmberEngine:AllPingPong:generate:59: rank=87 size=128
480620578389:0:89:EmberEngine:AllPingPong:generate:59: rank=89 size=128
480620578389:0:91:EmberEngine:AllPingPong:generate:59: rank=91 size=128
480620578389:0:93:EmberEngine:AllPingPong:generate:59: rank=93 size=128
480620578389:0:95:EmberEngine:AllPingPong:generate:59: rank=95 size=128
AllPingPong: total time 162382.900 us, loop 325, bufLen 20510, latency 249.820 u
s. bandwidth 0.082099 GB/s
480622985893:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=128
480622985893:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=128
480622985893:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=128
480622985893:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=128
480622985893:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=128
480622985893:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=128
480622985893:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=128
480622985893:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=128
480622985893:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=128
480622985893:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=128
480622985893:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=128
480622985893:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=128
480622985893:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=128
480622985893:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=128
480622985893:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=128
480622985893:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=128
480623501767:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=128
480623501767:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=128
480623501767:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=128
480623501767:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=128
480623501767:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=128
480623501767:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=128
480623501767:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=128
480623501767:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=128
480623501767:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=128
480623501767:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=128
480623501767:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=128
480623501767:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=128
480623501767:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=128
480623501767:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=128
480623501767:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=128
480623501767:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=128
AllPingPong: total time 155850.170 us, loop 325, bufLen 9000, latency 239.769 us
. bandwidth 0.037536 GB/s
Allreduce: ranks 128, loop 40, 1 double(s), latency 2464.302 us
Barrier: ranks 128, loop 15, latency 1405.482 us
Simulation is complete, simulated time: 756.147 ms
=== MiniMD 3D torus 4x4x8 (PPN=2) ===
=====
Topology          : 3D Torus 4x4x8 - MiniMD
Routers           : 128
Nodes per router  : 2 (local_ports)
Total nodes       : 256

```

Total MPI ranks : 256
PPN (ranks/node) : 2 <-- should be 2
Stats CSV : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/M
iniMD/Torus/PPN_2/results/minimd_torus_4x4x8_ppn2_nodes256_stats.csv

```
=====
180000:0:255:EmberEngine:AllPingPong:generate:59: rank=255 size=256
180000:0:254:EmberEngine:AllPingPong:generate:59: rank=254 size=256
180000:0:253:EmberEngine:AllPingPong:generate:59: rank=253 size=256
180000:0:252:EmberEngine:AllPingPong:generate:59: rank=252 size=256
180000:0:251:EmberEngine:AllPingPong:generate:59: rank=251 size=256
180000:0:250:EmberEngine:AllPingPong:generate:59: rank=250 size=256
180000:0:249:EmberEngine:AllPingPong:generate:59: rank=249 size=256
180000:0:248:EmberEngine:AllPingPong:generate:59: rank=248 size=256
180000:0:247:EmberEngine:AllPingPong:generate:59: rank=247 size=256
180000:0:246:EmberEngine:AllPingPong:generate:59: rank=246 size=256
180000:0:245:EmberEngine:AllPingPong:generate:59: rank=245 size=256
180000:0:244:EmberEngine:AllPingPong:generate:59: rank=244 size=256
180000:0:243:EmberEngine:AllPingPong:generate:59: rank=243 size=256
180000:0:242:EmberEngine:AllPingPong:generate:59: rank=242 size=256
180000:0:241:EmberEngine:AllPingPong:generate:59: rank=241 size=256
180000:0:240:EmberEngine:AllPingPong:generate:59: rank=240 size=256
180000:0:239:EmberEngine:AllPingPong:generate:59: rank=239 size=256
180000:0:238:EmberEngine:AllPingPong:generate:59: rank=238 size=256
180000:0:237:EmberEngine:AllPingPong:generate:59: rank=237 size=256
180000:0:236:EmberEngine:AllPingPong:generate:59: rank=236 size=256
180000:0:235:EmberEngine:AllPingPong:generate:59: rank=235 size=256
180000:0:234:EmberEngine:AllPingPong:generate:59: rank=234 size=256
180000:0:233:EmberEngine:AllPingPong:generate:59: rank=233 size=256
180000:0:232:EmberEngine:AllPingPong:generate:59: rank=232 size=256
180000:0:231:EmberEngine:AllPingPong:generate:59: rank=231 size=256
180000:0:230:EmberEngine:AllPingPong:generate:59: rank=230 size=256
180000:0:229:EmberEngine:AllPingPong:generate:59: rank=229 size=256
180000:0:228:EmberEngine:AllPingPong:generate:59: rank=228 size=256
180000:0:227:EmberEngine:AllPingPong:generate:59: rank=227 size=256
180000:0:226:EmberEngine:AllPingPong:generate:59: rank=226 size=256
180000:0:225:EmberEngine:AllPingPong:generate:59: rank=225 size=256
180000:0:224:EmberEngine:AllPingPong:generate:59: rank=224 size=256
180000:0:223:EmberEngine:AllPingPong:generate:59: rank=223 size=256
180000:0:222:EmberEngine:AllPingPong:generate:59: rank=222 size=256
180000:0:221:EmberEngine:AllPingPong:generate:59: rank=221 size=256
180000:0:220:EmberEngine:AllPingPong:generate:59: rank=220 size=256
180000:0:219:EmberEngine:AllPingPong:generate:59: rank=219 size=256
180000:0:218:EmberEngine:AllPingPong:generate:59: rank=218 size=256
180000:0:217:EmberEngine:AllPingPong:generate:59: rank=217 size=256
180000:0:216:EmberEngine:AllPingPong:generate:59: rank=216 size=256
180000:0:215:EmberEngine:AllPingPong:generate:59: rank=215 size=256
180000:0:214:EmberEngine:AllPingPong:generate:59: rank=214 size=256
180000:0:213:EmberEngine:AllPingPong:generate:59: rank=213 size=256
180000:0:212:EmberEngine:AllPingPong:generate:59: rank=212 size=256
180000:0:211:EmberEngine:AllPingPong:generate:59: rank=211 size=256
180000:0:210:EmberEngine:AllPingPong:generate:59: rank=210 size=256
180000:0:209:EmberEngine:AllPingPong:generate:59: rank=209 size=256
180000:0:208:EmberEngine:AllPingPong:generate:59: rank=208 size=256
180000:0:207:EmberEngine:AllPingPong:generate:59: rank=207 size=256
180000:0:206:EmberEngine:AllPingPong:generate:59: rank=206 size=256
180000:0:205:EmberEngine:AllPingPong:generate:59: rank=205 size=256
180000:0:204:EmberEngine:AllPingPong:generate:59: rank=204 size=256
180000:0:203:EmberEngine:AllPingPong:generate:59: rank=203 size=256
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[illegible]

162287341314:0:150:EmberEngine:AllPingPong:generate:59: rank=150 size=256
162287341314:0:152:EmberEngine:AllPingPong:generate:59: rank=152 size=256
162287341314:0:154:EmberEngine:AllPingPong:generate:59: rank=154 size=256
162287341314:0:156:EmberEngine:AllPingPong:generate:59: rank=156 size=256
162287341314:0:158:EmberEngine:AllPingPong:generate:59: rank=158 size=256
AllPingPong: total time 162289.951 us, loop 325, bufLen 20510, latency 249.677 u
s. bandwidth 0.082146 GB/s
162290131730:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=256
162290131730:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=256
162290131730:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=256
162290131730:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=256
162290131730:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=256
162290131730:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=256
162290131730:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=256
162290131730:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=256
162290131730:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=256
162290131730:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=256
162290131730:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=256
162290131730:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=256
162290131730:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=256
162290131730:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=256
162290131730:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=256
162290131730:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=256
162292109030:0:129:EmberEngine:AllPingPong:generate:59: rank=129 size=256
162292109030:0:131:EmberEngine:AllPingPong:generate:59: rank=131 size=256
162292109030:0:133:EmberEngine:AllPingPong:generate:59: rank=133 size=256
162292109030:0:135:EmberEngine:AllPingPong:generate:59: rank=135 size=256
162292109030:0:137:EmberEngine:AllPingPong:generate:59: rank=137 size=256
162292109030:0:139:EmberEngine:AllPingPong:generate:59: rank=139 size=256
162292109030:0:141:EmberEngine:AllPingPong:generate:59: rank=141 size=256
162292109030:0:143:EmberEngine:AllPingPong:generate:59: rank=143 size=256
162292109030:0:145:EmberEngine:AllPingPong:generate:59: rank=145 size=256
162292109030:0:147:EmberEngine:AllPingPong:generate:59: rank=147 size=256
162292109030:0:149:EmberEngine:AllPingPong:generate:59: rank=149 size=256
162292109030:0:151:EmberEngine:AllPingPong:generate:59: rank=151 size=256
162292109030:0:153:EmberEngine:AllPingPong:generate:59: rank=153 size=256
162292109030:0:155:EmberEngine:AllPingPong:generate:59: rank=155 size=256
162292109030:0:157:EmberEngine:AllPingPong:generate:59: rank=157 size=256
162292109030:0:159:EmberEngine:AllPingPong:generate:59: rank=159 size=256
162294899446:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=256
162294899446:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=256
162294899446:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=256
162294899446:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=256
162294899446:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=256
162294899446:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=256
162294899446:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=256
162294899446:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=256
162294899446:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=256
162294899446:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=256
162294899446:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=256
162294899446:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=256
162294899446:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=256
162294899446:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=256
162294899446:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=256
162294899446:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=256
317720265589:0:224:EmberEngine:AllPingPong:generate:59: rank=224 size=256
317720265589:0:226:EmberEngine:AllPingPong:generate:59: rank=226 size=256
317720265589:0:228:EmberEngine:AllPingPong:generate:59: rank=228 size=256

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318112273590:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=256
318112273590:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=256
318112273590:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=256
318112273590:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=256
318112273590:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=256
318112273590:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=256
318112273590:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=256
318112273590:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=256
318112273590:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=256
318112273590:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=256
318112273590:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=256
318112273590:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=256
318112273590:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=256
318112273590:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=256
318112273590:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=256
318114480598:0:128:EmberEngine:AllPingPong:generate:59: rank=128 size=256
318114480598:0:130:EmberEngine:AllPingPong:generate:59: rank=130 size=256
318114480598:0:132:EmberEngine:AllPingPong:generate:59: rank=132 size=256
318114480598:0:134:EmberEngine:AllPingPong:generate:59: rank=134 size=256
318114480598:0:136:EmberEngine:AllPingPong:generate:59: rank=136 size=256
318114480598:0:138:EmberEngine:AllPingPong:generate:59: rank=138 size=256
318114480598:0:140:EmberEngine:AllPingPong:generate:59: rank=140 size=256
318114480598:0:142:EmberEngine:AllPingPong:generate:59: rank=142 size=256
318114480598:0:144:EmberEngine:AllPingPong:generate:59: rank=144 size=256
318114480598:0:146:EmberEngine:AllPingPong:generate:59: rank=146 size=256
318114480598:0:148:EmberEngine:AllPingPong:generate:59: rank=148 size=256
318114480598:0:150:EmberEngine:AllPingPong:generate:59: rank=150 size=256
318114480598:0:152:EmberEngine:AllPingPong:generate:59: rank=152 size=256
318114480598:0:154:EmberEngine:AllPingPong:generate:59: rank=154 size=256
318114480598:0:156:EmberEngine:AllPingPong:generate:59: rank=156 size=256
318114480598:0:158:EmberEngine:AllPingPong:generate:59: rank=158 size=256
AllPingPong: total time 155825.891 us, loop 325, bufLen 9000, latency 239.732 us
. bandwidth 0.037542 GB/s
318116022797:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=256
318116022797:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=256
318116022797:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=256
318116022797:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=256
318116022797:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=256
318116022797:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=256
318116022797:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=256
318116022797:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=256
318116022797:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=256
318116022797:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=256
318116022797:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=256
318116022797:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=256
318116022797:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=256
318116022797:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=256
318116022797:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=256
318116022797:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=256
318119248314:0:129:EmberEngine:AllPingPong:generate:59: rank=129 size=256
318119248314:0:131:EmberEngine:AllPingPong:generate:59: rank=131 size=256
318119248314:0:133:EmberEngine:AllPingPong:generate:59: rank=133 size=256
318119248314:0:135:EmberEngine:AllPingPong:generate:59: rank=135 size=256
318119248314:0:137:EmberEngine:AllPingPong:generate:59: rank=137 size=256
318119248314:0:139:EmberEngine:AllPingPong:generate:59: rank=139 size=256
318119248314:0:141:EmberEngine:AllPingPong:generate:59: rank=141 size=256
318119248314:0:143:EmberEngine:AllPingPong:generate:59: rank=143 size=256
318119248314:0:145:EmberEngine:AllPingPong:generate:59: rank=145 size=256

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s. bandwidth 0.082274 GB/s
480154027585:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=256
480154027585:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=256
480154027585:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=256
480154027585:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=256
480154027585:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=256
480154027585:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=256
480154027585:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=256
480154027585:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=256
480154027585:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=256
480154027585:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=256
480154027585:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=256
480154027585:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=256
480154027585:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=256
480154027585:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=256
480154027585:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=256
480154027585:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=256
480625633582:0:129:EmberEngine:AllPingPong:generate:59: rank=129 size=256
480625633582:0:131:EmberEngine:AllPingPong:generate:59: rank=131 size=256
480625633582:0:133:EmberEngine:AllPingPong:generate:59: rank=133 size=256
480625633582:0:135:EmberEngine:AllPingPong:generate:59: rank=135 size=256
480625633582:0:137:EmberEngine:AllPingPong:generate:59: rank=137 size=256
480625633582:0:139:EmberEngine:AllPingPong:generate:59: rank=139 size=256
480625633582:0:141:EmberEngine:AllPingPong:generate:59: rank=141 size=256
480625633582:0:143:EmberEngine:AllPingPong:generate:59: rank=143 size=256
480625633582:0:145:EmberEngine:AllPingPong:generate:59: rank=145 size=256
480625633582:0:147:EmberEngine:AllPingPong:generate:59: rank=147 size=256
480625633582:0:149:EmberEngine:AllPingPong:generate:59: rank=149 size=256
480625633582:0:151:EmberEngine:AllPingPong:generate:59: rank=151 size=256
480625633582:0:153:EmberEngine:AllPingPong:generate:59: rank=153 size=256
480625633582:0:155:EmberEngine:AllPingPong:generate:59: rank=155 size=256
480625633582:0:157:EmberEngine:AllPingPong:generate:59: rank=157 size=256
480625633582:0:159:EmberEngine:AllPingPong:generate:59: rank=159 size=256
480628396752:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=256
480628396752:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=256
480628396752:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=256
480628396752:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=256
480628396752:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=256
480628396752:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=256
480628396752:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=256
480628396752:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=256
480628396752:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=256
480628396752:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=256
480628396752:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=256
480628396752:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=256
480628396752:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=256
480628396752:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=256
480628396752:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=256
480628396752:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=256
AllPingPong: total time 155881.936 us, loop 325, bufLen 9000, latency 239.818 us
. bandwidth 0.037528 GB/s
Allreduce: ranks 256, loop 40, 1 double(s), latency 2477.633 us
Barrier: ranks 256, loop 15, latency 1408.212 us
Simulation is complete, simulated time: 756.288 ms
=== MiniMD 3D torus 8x8x8 (PPN=2) ===
=====
Topology          : 3D Torus 8x8x8 - MiniMD
Routers           : 512

```

Nodes per router : 2 (local_ports)
Total nodes : 1024
Total MPI ranks : 1024
PPN (ranks/node) : 2 <-- should be 2
Stats CSV : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/M
iniMD/Torus/PPN_2/results/minimd_torus_8x8x8_ppn2_nodes1024_stats.csv
=====

180000:0:1023:EmberEngine:AllPingPong:generate:59: rank=1023 size=1024
180000:0:1022:EmberEngine:AllPingPong:generate:59: rank=1022 size=1024
180000:0:1021:EmberEngine:AllPingPong:generate:59: rank=1021 size=1024
180000:0:1020:EmberEngine:AllPingPong:generate:59: rank=1020 size=1024
180000:0:1019:EmberEngine:AllPingPong:generate:59: rank=1019 size=1024
180000:0:1018:EmberEngine:AllPingPong:generate:59: rank=1018 size=1024
180000:0:1017:EmberEngine:AllPingPong:generate:59: rank=1017 size=1024
180000:0:1016:EmberEngine:AllPingPong:generate:59: rank=1016 size=1024
180000:0:1015:EmberEngine:AllPingPong:generate:59: rank=1015 size=1024
180000:0:1014:EmberEngine:AllPingPong:generate:59: rank=1014 size=1024
180000:0:1013:EmberEngine:AllPingPong:generate:59: rank=1013 size=1024
180000:0:1012:EmberEngine:AllPingPong:generate:59: rank=1012 size=1024
180000:0:1011:EmberEngine:AllPingPong:generate:59: rank=1011 size=1024
180000:0:1010:EmberEngine:AllPingPong:generate:59: rank=1010 size=1024
180000:0:1009:EmberEngine:AllPingPong:generate:59: rank=1009 size=1024
180000:0:1008:EmberEngine:AllPingPong:generate:59: rank=1008 size=1024
180000:0:1007:EmberEngine:AllPingPong:generate:59: rank=1007 size=1024
180000:0:1006:EmberEngine:AllPingPong:generate:59: rank=1006 size=1024
180000:0:1005:EmberEngine:AllPingPong:generate:59: rank=1005 size=1024
180000:0:1004:EmberEngine:AllPingPong:generate:59: rank=1004 size=1024
180000:0:1003:EmberEngine:AllPingPong:generate:59: rank=1003 size=1024
180000:0:1002:EmberEngine:AllPingPong:generate:59: rank=1002 size=1024
180000:0:1001:EmberEngine:AllPingPong:generate:59: rank=1001 size=1024
180000:0:1000:EmberEngine:AllPingPong:generate:59: rank=1000 size=1024
180000:0:999:EmberEngine:AllPingPong:generate:59: rank=999 size=1024
180000:0:998:EmberEngine:AllPingPong:generate:59: rank=998 size=1024
180000:0:997:EmberEngine:AllPingPong:generate:59: rank=997 size=1024
180000:0:996:EmberEngine:AllPingPong:generate:59: rank=996 size=1024
180000:0:995:EmberEngine:AllPingPong:generate:59: rank=995 size=1024
180000:0:994:EmberEngine:AllPingPong:generate:59: rank=994 size=1024
180000:0:993:EmberEngine:AllPingPong:generate:59: rank=993 size=1024
180000:0:992:EmberEngine:AllPingPong:generate:59: rank=992 size=1024
180000:0:991:EmberEngine:AllPingPong:generate:59: rank=991 size=1024
180000:0:990:EmberEngine:AllPingPong:generate:59: rank=990 size=1024
180000:0:989:EmberEngine:AllPingPong:generate:59: rank=989 size=1024
180000:0:988:EmberEngine:AllPingPong:generate:59: rank=988 size=1024
180000:0:987:EmberEngine:AllPingPong:generate:59: rank=987 size=1024
180000:0:986:EmberEngine:AllPingPong:generate:59: rank=986 size=1024
180000:0:985:EmberEngine:AllPingPong:generate:59: rank=985 size=1024
180000:0:984:EmberEngine:AllPingPong:generate:59: rank=984 size=1024
180000:0:983:EmberEngine:AllPingPong:generate:59: rank=983 size=1024
180000:0:982:EmberEngine:AllPingPong:generate:59: rank=982 size=1024
180000:0:981:EmberEngine:AllPingPong:generate:59: rank=981 size=1024
180000:0:980:EmberEngine:AllPingPong:generate:59: rank=980 size=1024
180000:0:979:EmberEngine:AllPingPong:generate:59: rank=979 size=1024
180000:0:978:EmberEngine:AllPingPong:generate:59: rank=978 size=1024
180000:0:977:EmberEngine:AllPingPong:generate:59: rank=977 size=1024
180000:0:976:EmberEngine:AllPingPong:generate:59: rank=976 size=1024
180000:0:975:EmberEngine:AllPingPong:generate:59: rank=975 size=1024
180000:0:974:EmberEngine:AllPingPong:generate:59: rank=974 size=1024
180000:0:973:EmberEngine:AllPingPong:generate:59: rank=973 size=1024

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162287341314:0:626:EmberEngine:AllPingPong:generate:59: rank=626 size=1024
162287341314:0:628:EmberEngine:AllPingPong:generate:59: rank=628 size=1024
162287341314:0:630:EmberEngine:AllPingPong:generate:59: rank=630 size=1024
162287341314:0:632:EmberEngine:AllPingPong:generate:59: rank=632 size=1024
162287341314:0:634:EmberEngine:AllPingPong:generate:59: rank=634 size=1024
162287341314:0:636:EmberEngine:AllPingPong:generate:59: rank=636 size=1024
162287341314:0:638:EmberEngine:AllPingPong:generate:59: rank=638 size=1024
AllPingPong: total time 162289.951 us, loop 325, bufLen 20510, latency 249.677 u
s. bandwidth 0.082146 GB/s
162290131730:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=1024
162290131730:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=1024
162290131730:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=1024
162290131730:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=1024
162290131730:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=1024
162290131730:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=1024
162290131730:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=1024
162290131730:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=1024
162290131730:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=1024
162290131730:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=1024
162290131730:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=1024
162290131730:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=1024
162290131730:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=1024
162290131730:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=1024
162290131730:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=1024
162290131730:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=1024
162290131730:0:32:EmberEngine:AllPingPong:generate:59: rank=32 size=1024
162290131730:0:34:EmberEngine:AllPingPong:generate:59: rank=34 size=1024
162290131730:0:36:EmberEngine:AllPingPong:generate:59: rank=36 size=1024
162290131730:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=1024
162290131730:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=1024
162290131730:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=1024
162290131730:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=1024
162290131730:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=1024
162290131730:0:48:EmberEngine:AllPingPong:generate:59: rank=48 size=1024
162290131730:0:50:EmberEngine:AllPingPong:generate:59: rank=50 size=1024
162290131730:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=1024
162290131730:0:54:EmberEngine:AllPingPong:generate:59: rank=54 size=1024
162290131730:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=1024
162290131730:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=1024
162290131730:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=1024
162290131730:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=1024
162290131730:0:64:EmberEngine:AllPingPong:generate:59: rank=64 size=1024
162290131730:0:66:EmberEngine:AllPingPong:generate:59: rank=66 size=1024
162290131730:0:68:EmberEngine:AllPingPong:generate:59: rank=68 size=1024
162290131730:0:70:EmberEngine:AllPingPong:generate:59: rank=70 size=1024
162290131730:0:72:EmberEngine:AllPingPong:generate:59: rank=72 size=1024
162290131730:0:74:EmberEngine:AllPingPong:generate:59: rank=74 size=1024
162290131730:0:76:EmberEngine:AllPingPong:generate:59: rank=76 size=1024
162290131730:0:78:EmberEngine:AllPingPong:generate:59: rank=78 size=1024
162290131730:0:80:EmberEngine:AllPingPong:generate:59: rank=80 size=1024
162290131730:0:82:EmberEngine:AllPingPong:generate:59: rank=82 size=1024
162290131730:0:84:EmberEngine:AllPingPong:generate:59: rank=84 size=1024
162290131730:0:86:EmberEngine:AllPingPong:generate:59: rank=86 size=1024
162290131730:0:88:EmberEngine:AllPingPong:generate:59: rank=88 size=1024
162290131730:0:90:EmberEngine:AllPingPong:generate:59: rank=90 size=1024
162290131730:0:92:EmberEngine:AllPingPong:generate:59: rank=92 size=1024
162290131730:0:94:EmberEngine:AllPingPong:generate:59: rank=94 size=1024
162290131730:0:96:EmberEngine:AllPingPong:generate:59: rank=96 size=1024

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480151237169:0:582:EmberEngine:AllPingPong:generate:59: rank=582 size=1024
480151237169:0:584:EmberEngine:AllPingPong:generate:59: rank=584 size=1024
480151237169:0:586:EmberEngine:AllPingPong:generate:59: rank=586 size=1024
480151237169:0:588:EmberEngine:AllPingPong:generate:59: rank=588 size=1024
480151237169:0:590:EmberEngine:AllPingPong:generate:59: rank=590 size=1024
480151237169:0:592:EmberEngine:AllPingPong:generate:59: rank=592 size=1024
480151237169:0:594:EmberEngine:AllPingPong:generate:59: rank=594 size=1024
480151237169:0:596:EmberEngine:AllPingPong:generate:59: rank=596 size=1024
480151237169:0:598:EmberEngine:AllPingPong:generate:59: rank=598 size=1024
480151237169:0:600:EmberEngine:AllPingPong:generate:59: rank=600 size=1024
480151237169:0:602:EmberEngine:AllPingPong:generate:59: rank=602 size=1024
480151237169:0:604:EmberEngine:AllPingPong:generate:59: rank=604 size=1024
480151237169:0:606:EmberEngine:AllPingPong:generate:59: rank=606 size=1024
480151237169:0:608:EmberEngine:AllPingPong:generate:59: rank=608 size=1024
480151237169:0:610:EmberEngine:AllPingPong:generate:59: rank=610 size=1024
480151237169:0:612:EmberEngine:AllPingPong:generate:59: rank=612 size=1024
480151237169:0:614:EmberEngine:AllPingPong:generate:59: rank=614 size=1024
480151237169:0:616:EmberEngine:AllPingPong:generate:59: rank=616 size=1024
480151237169:0:618:EmberEngine:AllPingPong:generate:59: rank=618 size=1024
480151237169:0:620:EmberEngine:AllPingPong:generate:59: rank=620 size=1024
480151237169:0:622:EmberEngine:AllPingPong:generate:59: rank=622 size=1024
480151237169:0:624:EmberEngine:AllPingPong:generate:59: rank=624 size=1024
480151237169:0:626:EmberEngine:AllPingPong:generate:59: rank=626 size=1024
480151237169:0:628:EmberEngine:AllPingPong:generate:59: rank=628 size=1024
480151237169:0:630:EmberEngine:AllPingPong:generate:59: rank=630 size=1024
480151237169:0:632:EmberEngine:AllPingPong:generate:59: rank=632 size=1024
480151237169:0:634:EmberEngine:AllPingPong:generate:59: rank=634 size=1024
480151237169:0:636:EmberEngine:AllPingPong:generate:59: rank=636 size=1024
480151237169:0:638:EmberEngine:AllPingPong:generate:59: rank=638 size=1024
AllPingPong: total time 162038.005 us, loop 325, bufLen 20510, latency 249.289 u
s. bandwidth 0.082274 GB/s
480154027585:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=1024
480154027585:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=1024
480154027585:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=1024
480154027585:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=1024
480154027585:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=1024
480154027585:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=1024
480154027585:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=1024
480154027585:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=1024
480154027585:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=1024
480154027585:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=1024
480154027585:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=1024
480154027585:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=1024
480154027585:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=1024
480154027585:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=1024
480154027585:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=1024
480154027585:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=1024
480154027585:0:32:EmberEngine:AllPingPong:generate:59: rank=32 size=1024
480154027585:0:34:EmberEngine:AllPingPong:generate:59: rank=34 size=1024
480154027585:0:36:EmberEngine:AllPingPong:generate:59: rank=36 size=1024
480154027585:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=1024
480154027585:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=1024
480154027585:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=1024
480154027585:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=1024
480154027585:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=1024
480154027585:0:48:EmberEngine:AllPingPong:generate:59: rank=48 size=1024
480154027585:0:50:EmberEngine:AllPingPong:generate:59: rank=50 size=1024
480154027585:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=1024

[illegible]

[illegible]

480628396752:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=1024
480628396752:0:33:EmberEngine:AllPingPong:generate:59: rank=33 size=1024
480628396752:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=1024
480628396752:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=1024
480628396752:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=1024
480628396752:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=1024
480628396752:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=1024
480628396752:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=1024
480628396752:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=1024
480628396752:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=1024
480628396752:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=1024
480628396752:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=1024
480628396752:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=1024
480628396752:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=1024
480628396752:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=1024
480628396752:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=1024
480628396752:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=1024
480628396752:0:65:EmberEngine:AllPingPong:generate:59: rank=65 size=1024
480628396752:0:67:EmberEngine:AllPingPong:generate:59: rank=67 size=1024
480628396752:0:69:EmberEngine:AllPingPong:generate:59: rank=69 size=1024
480628396752:0:71:EmberEngine:AllPingPong:generate:59: rank=71 size=1024
480628396752:0:73:EmberEngine:AllPingPong:generate:59: rank=73 size=1024
480628396752:0:75:EmberEngine:AllPingPong:generate:59: rank=75 size=1024
480628396752:0:77:EmberEngine:AllPingPong:generate:59: rank=77 size=1024
480628396752:0:79:EmberEngine:AllPingPong:generate:59: rank=79 size=1024
480628396752:0:81:EmberEngine:AllPingPong:generate:59: rank=81 size=1024
480628396752:0:83:EmberEngine:AllPingPong:generate:59: rank=83 size=1024
480628396752:0:85:EmberEngine:AllPingPong:generate:59: rank=85 size=1024
480628396752:0:87:EmberEngine:AllPingPong:generate:59: rank=87 size=1024
480628396752:0:89:EmberEngine:AllPingPong:generate:59: rank=89 size=1024
480628396752:0:91:EmberEngine:AllPingPong:generate:59: rank=91 size=1024
480628396752:0:93:EmberEngine:AllPingPong:generate:59: rank=93 size=1024
480628396752:0:95:EmberEngine:AllPingPong:generate:59: rank=95 size=1024
480628396752:0:97:EmberEngine:AllPingPong:generate:59: rank=97 size=1024
480628396752:0:99:EmberEngine:AllPingPong:generate:59: rank=99 size=1024
480628396752:0:101:EmberEngine:AllPingPong:generate:59: rank=101 size=1024
480628396752:0:103:EmberEngine:AllPingPong:generate:59: rank=103 size=1024
480628396752:0:105:EmberEngine:AllPingPong:generate:59: rank=105 size=1024
480628396752:0:107:EmberEngine:AllPingPong:generate:59: rank=107 size=1024
480628396752:0:109:EmberEngine:AllPingPong:generate:59: rank=109 size=1024
480628396752:0:111:EmberEngine:AllPingPong:generate:59: rank=111 size=1024
480628396752:0:113:EmberEngine:AllPingPong:generate:59: rank=113 size=1024
480628396752:0:115:EmberEngine:AllPingPong:generate:59: rank=115 size=1024
480628396752:0:117:EmberEngine:AllPingPong:generate:59: rank=117 size=1024
480628396752:0:119:EmberEngine:AllPingPong:generate:59: rank=119 size=1024
480628396752:0:121:EmberEngine:AllPingPong:generate:59: rank=121 size=1024
480628396752:0:123:EmberEngine:AllPingPong:generate:59: rank=123 size=1024
480628396752:0:125:EmberEngine:AllPingPong:generate:59: rank=125 size=1024
480628396752:0:127:EmberEngine:AllPingPong:generate:59: rank=127 size=1024
AllPingPong: total time 155881.936 us, loop 325, bufLen 9000, latency 239.818 us
. bandwidth 0.037528 GB/s
Allreduce: ranks 1024, loop 40, 1 double(s), latency 2483.554 us
Barrier: ranks 1024, loop 15, latency 1414.420 us
Simulation is complete, simulated time: 756.628 ms
All PPN=2 MiniMD torus runs finished. Results in: /Users/stephenrajva/Downloads/
HPC-Upload 2/SST_Stimulator/MiniMD/Torus/PPN_2/results
-rw-r--r-- 1 stephenrajva staff 41777 28 Jun 07:03 results/minimd_torus_2x2
x2_ppn2_nodes16_stats.csv